Day - 25

GroupBy, Filtering & Sorting (Advanced Pandas) in Python

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Explanation

If Pandas is **Excel**, then:

Filtering = Excel Filters

Sorting = Excel Sort

GroupBy = Excel Pivot Table

Almost **80% of a Data Analyst's daily work** is:

"Filter the data → Group it → Sort it → Get insights"

Advanced Filtering in Pandas

Single Condition Filtering

```
df[df["City"] == "Mumbai"]
```

👉 All employees working in Mumbai

Numeric Filtering

```
df[df["Salary"] > 60000]
```

👉 High salary employees

Multiple Conditions (AND)

```
df[(df["City"] == "Mumbai") & (df["Salary"] > 60000)]
```

👉 Mumbai employees earning more than 60K

More Filtering Tricks

Multiple Conditions (OR)

```
df[(df["City"] == "Mumbai") | (df["City"] == "Pune")]
```

Using isin() (Very Useful)

```
df[df["Department"].isin(["IT", "Finance"])]
```

Filtering Using String Conditions

```
df[df["Employee_Name"].str.startswith("R")]
```

Sorting Data (Advanced Level)

✓ Sort by One Column

df.sort_values("Salary", ascending=False)

✓ Sort by Multiple Columns

df.sort_values(["Department", "Salary"], ascending=[True, False])

👉 Department-wise highest salary first

✓ Top N Records

df.sort_values("Salary", ascending=False).head(5)

✓ Bottom N Records

df.sort_values("Salary").head(5)

GroupBy

- “Split the data into groups → Apply calculation → Combine results”

- groupby() groups rows based on one or more columns and applies aggregation functions like:
  - sum
  - mean
  - min
  - max
  - count

GroupBy – Examples

Average Salary by Department

```
df.groupby("Department")["Salary"].mean()
```

Total Salary by City

```
df.groupby("City")["Salary"].sum()
```

Employee Count by Department

```
df.groupby("Department")["Employee_ID"].count()
```

GroupBy – Multiple Aggregations

```
df.groupby("Department")["Salary"].agg(["min", "max", "mean", "count"])
```

GroupBy on Multiple Columns

```
df.groupby(["Department", "City"])["Salary"].mean()
```

👉 Average salary per department per city

Sorting GroupBy Results

```
df.groupby("Department")["Salary"].mean().sort_values(ascending=False)
```

GroupBy + Filtering

✓ **Departments with Average Salary > 55,000**

```
dept_avg = df.groupby("Department")["Salary"].mean()
dept_avg[dept_avg > 55000]
```

Reset Index (Very Important for Export)

```
summary = df.groupby("City")["Salary"].sum().reset_index()
```